Unit: 6 Unit 2: Exponents Radicals **Name:** ________________
Topic: Exponents **Date:** ___________

Challenge (Habanero)

Q1. A gamer's score is 2^x points. If the score doubles, what is the new score in terms of x?
(A) 2^{x+1}
(B) 2^{x-1}
(C) $2x$
(D) $2^x + 2$
(E) 2^{2x}

Q2. A coding error in a game causes data usage to increase by a factor of 3^2. If the original data usage was x MB, what is the new data usage?
(A) $3x$
(B) $9x$
(C) x^3
(D) 3^x
(E) x^2

Q3. The number of players in an online game is 5^x. If 15 new players join, what is the new number of players in terms of x?
(A) 5^{x+1}
(B) $5^x + 15$
(C) 5^{x-1}
(D) $5x + 15$
(E) 15^x

Q4. A gaming platform's user base grows by a factor of 2^3 each year. If the current user base is x, what will it be in 2 years?
(A) 2^3x
(B) 2^6x
(C) $8x$
(D) x^2
(E) $2x$

Q5. The cost of in-game items is 3^x dollars. If the cost increases by a factor of 3, what is the new cost in terms of x?
(A) 3^{x+1}
(B) 3^{x-1}
(C) $3x$
(D) $3^x + 3$
(E) 9^x

Q6. The score of a game is calculated by the formula $2^x cdot 3^y$. If $x = 2$ and $y = 3$, what is the score?
(A) 72
(B) 108
(C) 144
(D) 216
(E) 288

Q7. The number of levels in a game is x^2 and the number of points required to pass each level is 2^x. What is the total number of points required to pass all levels?
(A) 2^{x^2}
(B) $x^2 cdot 2^x$
(C) $x cdot 2^{x^2}$
(D) $2x$
(E) $x^2 + 2^x$

Q8. The time taken to complete a level in a game is 3^x minutes. If the time taken is doubled, what is the new time in terms of x?
(A) $2 cdot 3^x$
(B) 3^{x+1}
(C) 3^{2x}
(D) $3^x + 2$
(E) 6^x

Open Ended Questions (FRQ)

Q9. A tech company's data storage capacity is 2^{10} GB. If the capacity is increased by a factor of 2^3, what is the new capacity in GB? Show your work and explain your reasoning.

Q10. A game developer wants to create a scoreboard that displays the top 5 players with the highest scores. If the scores are calculated by the formula $x^2 cdot 2^y$, where x is the number of levels completed and y is the number of points earned, what is the score of the player who completed 3 levels and earned 12 points? Show your work and explain your reasoning.

Chef's Tips

- Read the questions carefully
- Use the formula sheet provided
- Show all work and explain reasoning